

Diffie-Hellman Key Exchange

Textbook Chapter 10

Cryptography Roadmap

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	Symmetric-key	Asymmetric-key
Confidentiality	● One-time pads ● Block ciphers with chaining modes (e.g. AES-CBC)	● RSA encryption ● ElGamal encryption
Integrity, Authentication	● MACs (e.g. HMAC)	● Digital signatures (e.g. RSA signatures)

- Hash functions
- Pseudorandom number generators
- Public key exchange (e.g. Diffie-Hellman)
- Key management (certificates)
- Password management

Diffie-Hellman Key Exchange

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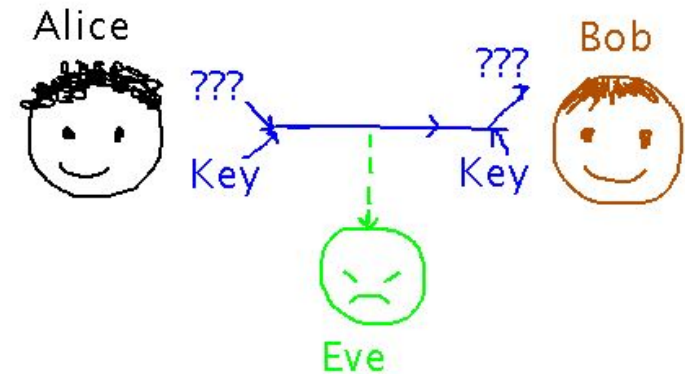
https://www.youtube.com/watch?v=YEBfamv-_do



Sharing Secret Keys

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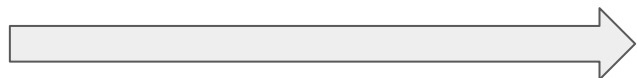
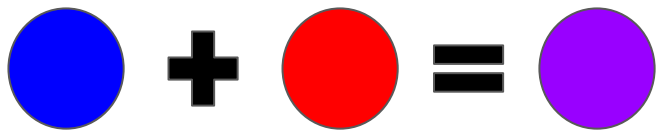
- Eve is always listening
- How would Alice and Bob share a secret key through an insecure channel?



Mixing Colors!

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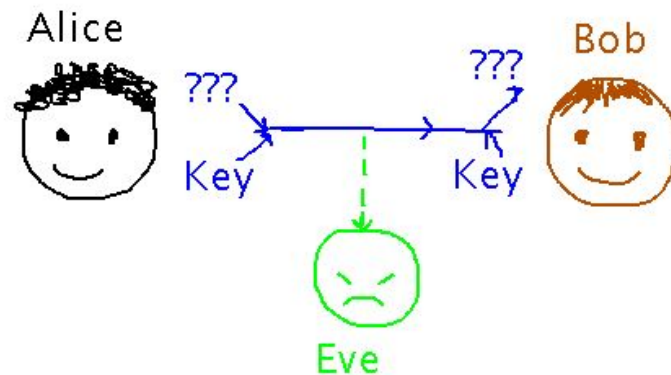
- Suppose the key is a secret color.
- Mixing colors is a one-way function



Easy



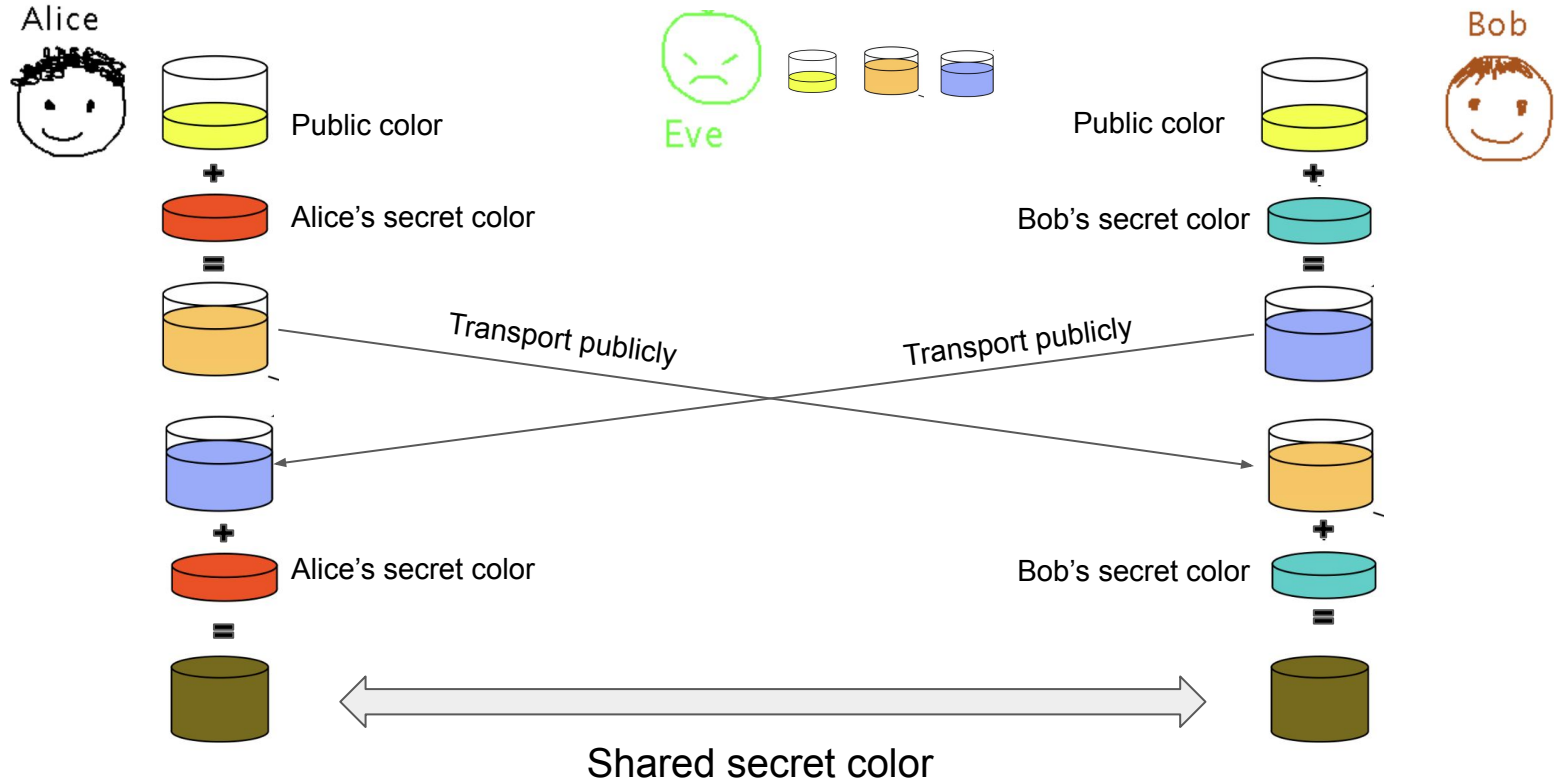
Hard



Mathematical equivalent of color mixing: Discrete logarithm problem

Secure Color Sharing

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Secure Color Sharing

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- Suppose Alice and Bob want a secret *paint color*, but Eve can see paint colors sent between Alice and Bob
- Alice generates a secret color **amber A**, and Bob generates a secret color **blue B**
- Alice and Bob agree on a common, public color **green G**
- They both mix their secret colors with **G**, so Alice has **green-amber GA**, and Bob has **green-blue GB**
- Alice sends **GA** to Bob, and Bob sends **GB** to Alice
 - Note: Eve now knows the colors **GA** and **GB**! Assume that it is hard to separate colors.
- Alice knows **GB**, so she can mix in **A** to form green-amber-blue **GAB**. Bob knows **GA**, so he can mix in **B** to form **GAB**, as well!
 - Eve only knows **G**, **GA**, and **GB**, so she can only form **green-amber-green-blue GAGB**, which is not the same!



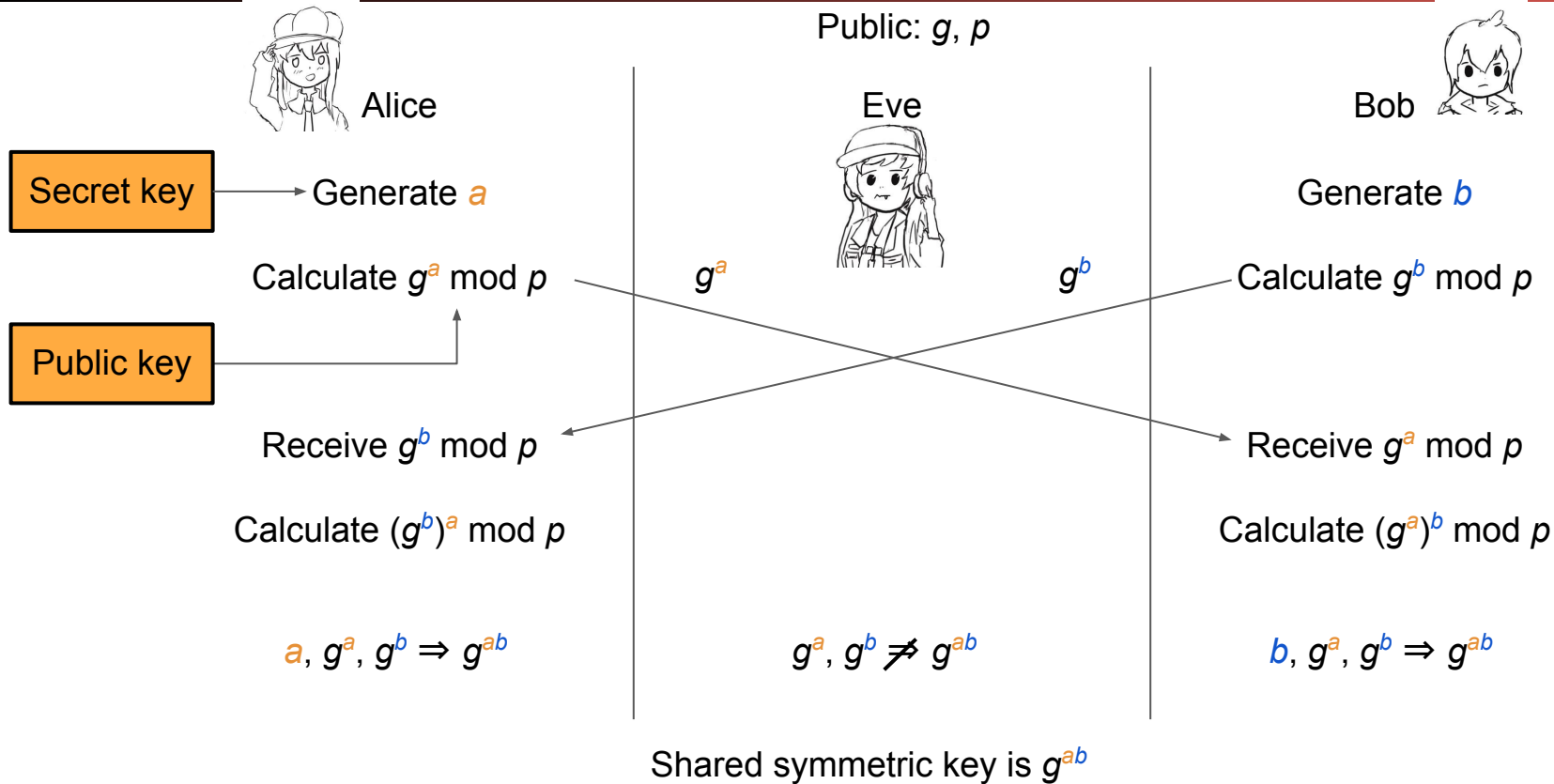
Discrete Log Problem and Diffie-Hellman Problem

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- There is a mathematical version of this
- Assume everyone knows a large prime p (e.g. 2048 bits long) and a number g
- **Discrete logarithm problem (discrete log problem):** Given $g, p, g^a \bmod p$ for random a , it is computationally hard to find a
- **Diffie-Hellman assumption:** Given $g, p, g^a \bmod p$, and $g^b \bmod p$ for random a, b , no polynomial time attacker can distinguish between a random value R and $g^{ab} \bmod p$.
 - Intuition: The best known algorithm is to first calculate a and then compute $(g^b)^a \bmod p$, but this requires solving the discrete log problem, which is hard!
 - Note: Multiplying the values doesn't work, since you get $g^{a+b} \bmod p \neq g^{ab} \bmod p$

Diffie-Hellman Key Exchange

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Discrete Log Problem and Diffie-Hellman Problem

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- g, p are publicly shared
- a, b are secret to Alice and Bob. Only Alice knows a , and only Bob knows b .
- $g^a \bmod p$, and $g^b \bmod p \Rightarrow$ these too are publicly shared.
- g is called the generator (also called primitive root mod p)
 - g lies within $[2, p-1]$
 - $g^a \bmod p$ distributes uniformly over $[1, p-1]$ for all possible choices of a .
 - In the context of number theory, if g is a primitive root of p , then $g^a \bmod p$ has residues $[1, p-1]$ and it has a period of $p-1$.
 - All possible choices of “ a ” includes: $[0, p-2]$. As soon as “ a ” becomes $p-1$, the modulo starts repeating from 1.
 - According to Euler’s theorem, $g^{\varphi(p)} \equiv 1 \bmod p$. Since p is a prime number, $\varphi(p) = p-1$.

Discrete Log Problem and Diffie-Hellman Problem

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In Diffie–Hellman key exchange, the values a and b (private keys) must satisfy these properties:

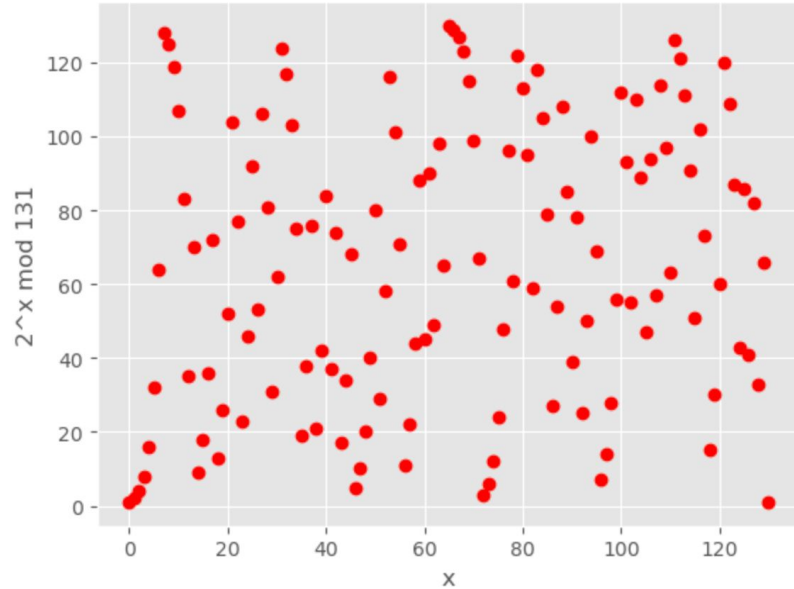
1. **Randomly chosen:** They should be chosen randomly and kept secret.
2. **In the range:** $1 \leq a, b \leq p - 2$, where p is a large prime.
3. **Relatively prime not required:** They do **not** need to be coprime to $p - 1$.
4. **Sufficiently large:** To ensure security, a and b should be large integers (hundreds or thousands of bits in real applications).

These private keys are then used to compute public keys:

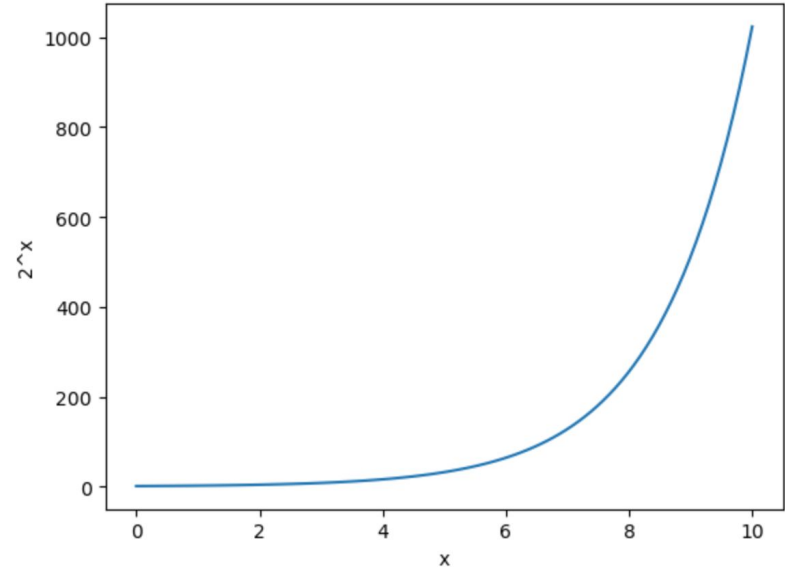
- $A = g^a \bmod p$
- $B = g^b \bmod p$

Discrete Log Problem

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Plot of $g^a \bmod p$



Plot of g^a

<https://www.desmos.com/calculator/3j0h11za2s>

Why must g be a primitive root?

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- If g isn't a generator (primitive root), then
 - $g^{ab} \bmod p$ only covers a subset of $[1, 2, \dots, p-1]$
 - The shared secret $g^{ab} \bmod p$ lives in that smaller subgroup, so it has less entropy
 - Less entropy = more predictable = easier to break / brute force.